

Scalable Portals for Distant Object Manipulation in Virtual Reality

Seungwoo Eun*
Pusan National University

Isaac Cho†
Utah State University

Myungho Lee‡
Pusan National University

ABSTRACT

Portal-based interaction supports distant object manipulation in Virtual Reality (VR) by bringing remote content into the user's reachable space, but its effectiveness is often limited by sensitivity to object scale. In this work, we examine scalable portal designs that reduce scale-related limitations in portal-based interaction. We explore two designs: a single scalable portal with explicit user control of scale and a dual-portal approach that decouples selection and manipulation through automatic scale normalization. A preliminary pilot user study using a 3D docking task suggests that both scalable portal designs reduce mean task completion time compared to a static portal baseline, particularly for smaller targets, while introducing modest selection overhead.

Index Terms: Virtual Reality, 3D User Interfaces, Scalable Interaction, 3D Manipulation

1 INTRODUCTION

Object manipulation in Virtual Reality (VR) is a fundamental task in 3D user interfaces (3DUI), often requiring users to interact with distant targets of varying sizes and distances [1, 2]. Portal-based interaction techniques [3, 4] address this by modulating the distance between the user and a remote target, enabling direct, simple hand-based manipulation without requiring physical navigation. By bringing distant content into the user's reachable space, portals preserve spatial awareness of the surrounding environment while supporting natural interaction. However, portal-based interaction remains sensitive to target scale. When a target is too small, precise selection and manipulation become difficult due to limited effective resolution within the portal. Conversely, overly large targets exceed the comfortable interaction space. To address these limitations, we propose a scalable portal approach that dynamically adjusts target scale based on interaction context. This design aims to support consistent manipulability across a wide range of object sizes while preserving the core benefits of portal-based object manipulation.

We utilize a World-in-Miniature (WiM) metaphor [5] to instantiate a portal that initially brings all target objects within the user's reachable workspace through the portal. This allows users to create a portal with an immediate overview and an accessible selection space without requiring navigation. Building on this, we designed two portal-based interaction approaches. The first approach (Figure 1b) is a single scalable portal in which the user dynamically adjusts the portal scale to balance overview and manipulation fidelity. While this approach offers the flexibility of scaling a portal, it requires the user to manage scale explicitly, introducing additional interaction overhead. The second approach is a dual-portal design (Figure 1c), separating selection and manipulation into two portals that present the object at an ergonomically optimized scale [6].

We conducted a pilot study to compare the proposed scalable and dual-portal approaches against a single non-scalable portal baseline

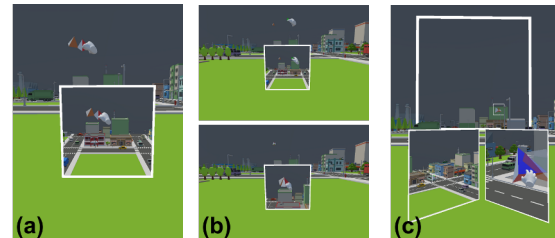


Figure 1: The evaluated portal techniques. (a) Static Portal, (b) Scalable Portal, and (c) Dual-Portal.

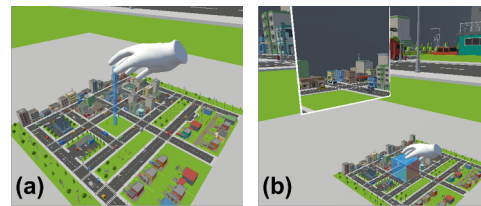


Figure 2: The WiM-Based Portal Creation Technique. (a) The user hovers the hand to the designated region of interest in the WiM. (b) Creation of the portal volume with a cube-based selection.

to evaluate how scale adjustability impacts remote object interaction performance. The results show that both the scalable and dual-portal approaches outperform the single-portal baseline by reducing overall task completion time, although they require additional time to initially engage the target object for manipulation.

2 SCALABLE PORTAL TECHNIQUES

Our scalable portal interaction follows a two-step process: (1) portal creation to define the spatial region of interest and (2) scalable portal interaction to support selection and manipulation within the instantiated portal.

WiM-based Portal Creation: We employed a WiM metaphor to create and place a portal in a virtual environment. The user first selects a region of interest by pointing a visual indicator over the corresponding area within the WiM (Figure 2a). After selecting a region of interest, the user defines a volumetric selection using a cube-based selection to specify the spatial extent of the portal (Figure 2b). During this process, a blue bounding box visualizes the portal's interaction volume and is scaled to ensure that all objects contained within it can be comfortably reached within the user's arm reach through the portal, while the red side indicates the portal's facing direction. The user can adjust the size of the portal volume by moving the hand, allowing the volume to be enlarged or reduced before instantiation. Once confirmed, the selected volume is instantiated as a portal positioned at a fixed, ergonomically reachable distance in front of the user.

Scalable Portal: The Scalable Portal technique provides a single portal through which the user adjusts the scale of the portal, both selection and manipulation (Figure 1b). The user can adjust the scale during interaction to balance global overview and local manipulation precision. Scaling is controlled via controller-based

*e-mail: eun.seungwoo@pusan.ac.kr

†e-mail: isaac.cho@usu.edu

‡e-mail: myungho.lee@pnu.edu

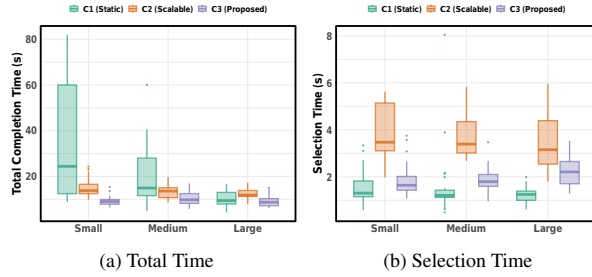


Figure 3: (a) Mean total completion time and (b) selection time by technique and target size.

rate control, allowing the user to zoom in to enlarge a small target for precise manipulation or zoom out to regain spatial context.

Dual-Portal: The Dual-Portal technique extends scalable portal interaction by instantiating additional portals based on object selection (Figure 1c). Selecting a target object and a docking object instantly triggers the creation of two object-specific manipulation portals, each automatically scaled and configured to look at the object’s bounding-box center, and positioned to keep the full object centered and visible in the portal view.

3 EXPERIMENT DESIGN

We conducted a preliminary pilot user study approved by the Institutional Review Board of the authors’ university (No. 2025.166.HR). Six participants (5 male, 1 female; mean age = 26.8, SD = 2.4) were recruited, all of whom had prior experience with VR. The study was conducted using a Meta Quest 3 VR Head Mounted Display, and the environment was built in Unity 2020.3.42f1 using the XR Interaction Toolkit.

The virtual environment represents a large-scale outdoor scene (200m × 200m) consisting of multiple buildings and trees, within which a group of tetrahedra is placed for docking tasks (Figure 2).

Experimental Design: The study employed a 3 × 3 within-subjects design that manipulated interaction technique and target size. Three portal-based interaction techniques were evaluated: a *Static Portal* as a baseline, a *Scalable Portal* that allowed users to adjust portal scale, and a *Dual-Portal* that separated selection and manipulation while automatically normalizing object scale. Target size varied across *Small* (1m), *Medium* (2m), and *Large* (4m) conditions.

Task and Procedure: Participants performed a 3D docking task in which they aligned a movable colored-tetrahedron with a semi-transparent target. Position tolerance was set to 5% of the object’s physical size, while rotation tolerance (5°) and dwell time (0.5s) were held constant to isolate interaction efficiency. The order of interaction techniques was fully counterbalanced using a Latin square, and target sizes were randomized with each trial. Task completion time was separated into selection time and manipulation time, allowing us to examine how each interaction technique affected target acquisition and object alignment independently. WiM-based portal selection was excluded to focus the evaluation on the effects of portal scaling and manipulation design.

4 RESULTS AND DISCUSSION

Figure 3 summarizes the mean total task completion time and selection time across interaction techniques and target sizes. Overall, both the *Scalable Portal* and *Dual-Portal* show lower mean total completion times than the *Static Portal*, particularly for smaller targets. In contrast, the *Static Portal* exhibits a clear increase in total completion time as target size decreases, indicating reduced effectiveness when precise manipulation is required.

For the selection time, both *Scalable Portal* and *Dual-Portal* require longer mean selection time than *Static Portal*. This reflects the additional interaction needed to scale or instantiate the portal to engage with the target object. *Scalable Portal*, in particular, may have been sensitive to scaling parameters, potentially increasing selection time. However, this initial overhead was offset by substantially shorter manipulation times, especially for small targets, resulting in lower overall completion time.

Across target sizes, *Dual-Portal* demonstrates the most consistent performance, with stable mean completion time regardless of object size. This trend suggests that automatic portal scaling during manipulation reduces sensitivity to object size and mitigates the precision limitations observed with the static portal approach.

Overall, the observed mean differences consistently favor the proposed scalable portal techniques. These preliminary results suggest that scaling portals, particularly through separating selection and manipulation, can improve efficiency and robustness in distance object manipulation tasks.

5 CONCLUSION AND FUTURE WORK

In this work, we investigate scalable portal interactions as a means to address the scale sensitivity inherent in portal-based object manipulation. Building on a WiM-based portal for initial creation, we explore two complementary approaches: a single scalable portal that enables explicit user control over scale, and a dual-portal that decouples selection and manipulation through automatic scale normalization. A preliminary pilot study comparing these approaches against a static, non-scalable portal baseline indicates that both scalable designs reduce mean overall task completion time, particularly for smaller targets, despite introducing additional time to initially engage the target object.

Future work will evaluate additional metrics (e.g., errors, precision) to more fully characterize the trade-offs between single scalable and dual-portal designs. In addition, we will also investigate how different scale-control strategies and transition behaviors influence usability and workload across extended task sequences and more complex virtual environments.

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